

Study on decomposition and calculation method of EPS assist characteristic curve

Proc IMechE Part D:

J Automobile Engineering

1–10

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DOI: 10.1177/0954407020987062

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Abstract

Steering feel is closely related to the matching of the EPS assist characteristic curve, however, due to the lack of theoretical basis for the design of the EPS assist characteristic curve, the steering feel can only be changed indirectly by adjusting the magnitude of assist, which is very difficult. To control steering feel directly and reduce the difficulty of adjustment, this paper proposes a decomposition and calculation method of the EPS assist characteristic curve. At first, the mechanism of the EPS assist characteristic curve is revealed. It is found that the process of designing and adjusting the EPS assist characteristic curve is a process of changing the corresponding relationship between the steering wheel torque and the steering motion intensity based on considering vehicle dynamic characteristics. On this basis, the driver's desired steering motion intensity and the pinion angle position are taken as intermediate variables, the EPS assist characteristic curve is decomposed into three parts: driving style, steady-state inverse characteristics of chassis dynamics, and steady-state inverse characteristics of steering system dynamics. According to the designed driving style and the calibrated steady-state inverse characteristics of chassis dynamics and steering system dynamics, the EPS assist characteristic curve can be directly calculated. The test results show that the EPS system adopting assist characteristic curve calculated can realize the designed driving style and provide consistent and controllable steering feel on the premise of meeting the requirements of steering portability and road feel.

Keywords

Electric power steering, assist characteristic curve, driving style, steering feel, vehicle dynamic characteristics

Date received: 22 July 2020; accepted: 10 December 2020

Introduction

The electric power steering (EPS) system is powered by an engine-independent motor. Compared with the hydraulic power steering (HPS) system, it has the advantages of energy-saving, environment-friendly, and safety, and its assist torque can be adjusted by the electronic control unit. The EPS system can customize the steering feel. In recent years, the EPS system is gradually replacing the HPS system and has been widely used in automobiles.^{1,2} The difficulty of matching the EPS system with vehicle lies in the control of steering feel to make the driver's steering input and the vehicle's steering motion intensity have a satisfying fixed relationship.^{3,4} As the most basic function of the EPS system, assistant control has an essential impact on the performance of the EPS system. The EPS assist characteristic curve, which is the core of assistant control, has a significant effect on the steering portability and road

feel. An EPS system with proper assist characteristics can solve the contradiction between steering portability and road feel, and can also provide consistent and controllable steering feel.⁵ Therefore, it is critical to investigate the EPS assist characteristics curve.

At present, studies on the EPS system have focused on its control strategy,^{6–10} with fewer studies on the assist characteristics. Shi et al.¹¹ discussed several issues about the design of assist characteristics, gave the requirements for assist characteristics, and compared the advantages and disadvantages of linear, bilinear,

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and curvilinear assist characteristic curves, which has a guiding significance for the design of EPS characteristic curve. Liu¹² designed the linear EPS assist characteristic curve based on the setting desired relationship between steering wheel angle and steering wheel torque. Yue¹³ described a design method of linear EPS assist characteristic curve based on the driver's steering wheel effort preference under typical conditions. He et al.¹⁴ pointed out that the in-situ steering assist can be designed first, and then the speed coefficient can be adjusted, based on which the assist characteristic curve can be calculated. Chen et al.¹⁵ investigated the effect of the load of vehicle on the assist characteristic curve. Liu and Ji¹⁶ proposed a method for designing the EPS assist characteristic curve based on the inherent "road feel" of vehicle. Ren and Deng¹⁷ investigated the relationship between the ideal steering wheel effort characteristics and the EPS assist characteristic curve, and pointed out that the curve of assist characteristic should have concave feature. Zhao¹⁸ used a genetic algorithm to optimize the parameters of assist characteristic curve with the minimum actual steering wheel torque and ideal steering wheel torque deviation as the optimization goal. He and Xiao¹⁹ proposed a design method of linear EPS assist characteristic curve considering the influence of speed, steering wheel torque, and load, the assist gain corresponding to different vehicle speeds and loads was obtained by fuzzy rule. In summary, most of the current EPS assist characteristic curve design methods do not reveal the essence of the design, do not systematically consider the driver's steering intention, and do not consider the chassis dynamics and steering system dynamics separately. As a result, the designed EPS assist characteristic curve can only satisfy the driver's steering feel preference under some typical conditions, which increases the difficulty and workload of adjusting the steering feel in the real vehicle field test in the late stage of EPS development.

In this paper, the mechanism of the EPS assist characteristic curve is revealed, which provides a theoretical basis for its design and adjustment. A decomposition and calculation method of the EPS assist characteristic curve is proposed, which decouples the driver's steering intention and vehicle dynamics characteristics coupled in the EPS assist characteristic curve, and reduces the difficulty of controlling steering feel.

Mechanism analysis of the EPS assist characteristic curve

To facilitate the analysis of the mechanism of the EPS assist characteristic curve, the steering system model, the vehicle model, and the tire model are reasonably simplified.

The steering system is simplified as a single degree of freedom model, ignoring the dry friction of the steering system and only considering the motion of pinion. The inertia and damping of the entire steering system are

equivalent converted to those of the pinion, and the dynamics equation of the simplified steering system is

$$\begin{cases} J_{eq}\ddot{\delta}_p + C_{eq}\dot{\delta}_p = T_{sw} + T_a - T_p \\ T_{sw} = K_{tb}(\delta_{sw} - \delta_p) \\ T_a = K_a T_{sw} \end{cases} \quad (1)$$

where J_{eq} and C_{eq} are the moment of inertia and damping coefficient of the steering system, respectively, T_{sw} is the steering wheel torque, T_a and T_p are the assist torque and steering resistance torque, respectively, δ_{sw} and δ_p are the steering wheel angle and pinion angle, respectively, K_{tb} is the torsional stiffness of the torsion bar, K_a is assist gain.

The vehicle is simplified to a two degree of freedom model, only considering the lateral and yaw motion of the car, and the dynamics equation of simplified vehicle is

$$\begin{cases} k_1\left(\beta + \frac{a\omega_r}{v_x} - \delta_f\right) + k_2\left(\beta - \frac{b\omega_r}{v_x}\right) = m(\dot{v}_y + v_x\omega_r) \\ ak_1\left(\beta + \frac{a\omega_r}{v_x} - \delta_f\right) - bk_2\left(\beta - \frac{b\omega_r}{v_x}\right) = I_z\dot{\omega}_r \end{cases} \quad (2)$$

where k_1 and k_2 are front and rear tire cornering stiffness, respectively, a and b are the longitudinal distances from vehicle's center of mass to the front and rear axles, respectively, m is the vehicle's mass, δ_f is front wheel steering angle, v_x and v_y are the longitudinal and lateral velocity. β and ω_r are the sideslip angle and yaw rate of the vehicle's center of mass, respectively.

The tire is simplified to a linear model, and only considering the self-aligning torque caused by the reactive force of the front lateral force, as shown below is

$$T_r = 2\xi k_1\left(\beta + \frac{a\omega_r}{v_x} - \delta_f\right) \cong K_{eq}n_G^2\delta_f \quad (3)$$

where T_r is self-aligning torque, ξ is a lateral trail consisting of the pneumatic trail and the caster trail, K_{eq} is the tire stiffness equivalent to the pinion, and n_G is the steering gear ratio with the relation of $\delta_p = n_G\delta_f$.

At steady state, it can be known from equation (1) that the steering wheel angle and the pinion angle meet the relationship shown in the following equation.

$$\delta_p = \delta_{sw} - T_{sw}/K_{tb} \quad (4)$$

At steady state, $\dot{v}_y$ and $\dot{\omega}_r$ are zeros. Considering $a_y = v_x\omega_r$ and $\delta_p = n_G\delta_f$, it can be known from equation (2) that the lateral acceleration and the pinion angle meet the relationship shown in equation (5).

$$G_{a_y} = \frac{a_y}{\delta_p} \bigg|_s = \frac{v_x^2/L}{1 + Kv_x^2 n_G} \quad (5)$$

where a_y is the lateral acceleration of the vehicle's center of mass, K is the stability factor, L is the wheelbase.

It can be known from equation (3) that there is a definite relationship between the front wheel angle and self-aligning torque. Considering the determined transmission ratio relationship between the pinion rotation

angle and the front wheel rotation angle, and between the steering resistance torque and the self-aligning torque, it can be obtained that there is a definite correspondence between pinion angle and steering resistance torque, $T_p = K_{eq}\delta_p$. At steady state, steering wheel torque and motor assist torque are balanced with steering resistance torque, $T_{sw} + T_a = T_p$. Considering that the assist gain coefficient is a specific value, the steering resistance torque has a definite relation with the steering wheel torque, $T_{sw} = T_p/(1 + K_a)$.

In summary, at steady state, the steering wheel angle or lateral acceleration, which represents the intensity of the steering motion, is the constant value. The constant steering motion intensity corresponds to the determined pinion angle, and the determined pinion angle corresponds to the determined steering resistance torque. Considering that the assist gain coefficient, steering wheel torque is a specific value, as shown in Figure 1.

Decomposition and calculation method of the EPS assist characteristic curve

Based on the analysis of the EPS assist characteristic curve mechanism in Section 2, a decomposition and calculation method of the EPS assist characteristic curve is proposed. Taking the desired steering motion intensity and pinion angle as intermediate variables, the driver's steering intention and the vehicle dynamics

characteristics coupled in the EPS assist characteristic curve are decoupled. The EPS assist characteristic curve is decomposed into three parts: driving style, steady-state inverse characteristics of chassis dynamics, and steady-state inverse characteristics of steering system dynamics, as shown in Figure 2. Based on the designed driving style and the calibrated steady-state inverse characteristics of chassis dynamics and steering system dynamics, the EPS assist characteristic curve can be directly calculated.

Design of driving style

The driving style is the corresponding relation between the driver's steering input torque and the desired steering motion intensity. However, there is no unified international standard for the definition of steering motion intensity. This paper believes that the steering wheel angle or lateral acceleration represents the steering motion intensity and indicates the driver's steering intention. The corresponding relationship between the driver's steering input torque and the steering wheel angle or lateral acceleration represents the driving style of the vehicle. It was found that the steering dynamic response of the car was related to the longitudinal speed. With different longitudinal speeds, the variable representing the steering motion intensity changes, and the definition of the driving style also changes accordingly, as follows:

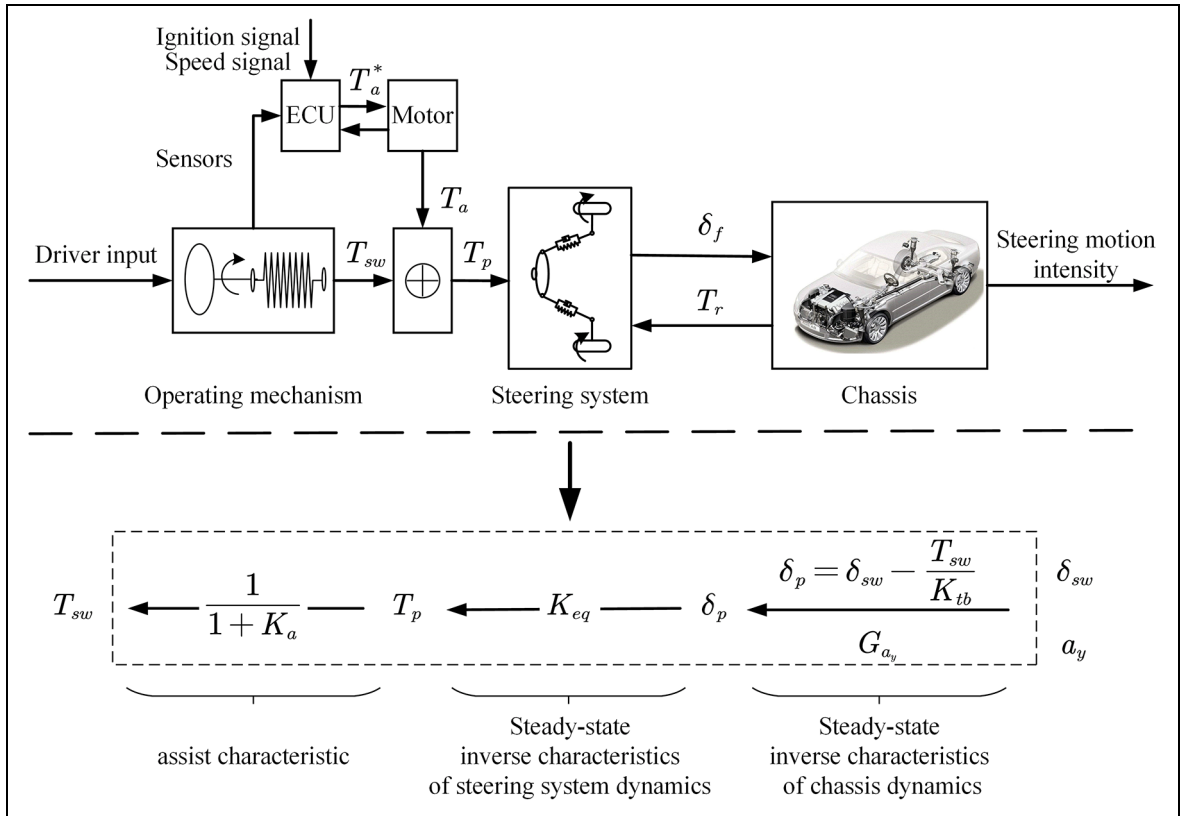


Figure 1. Schematic diagram of EPS assist characteristic curve mechanism.

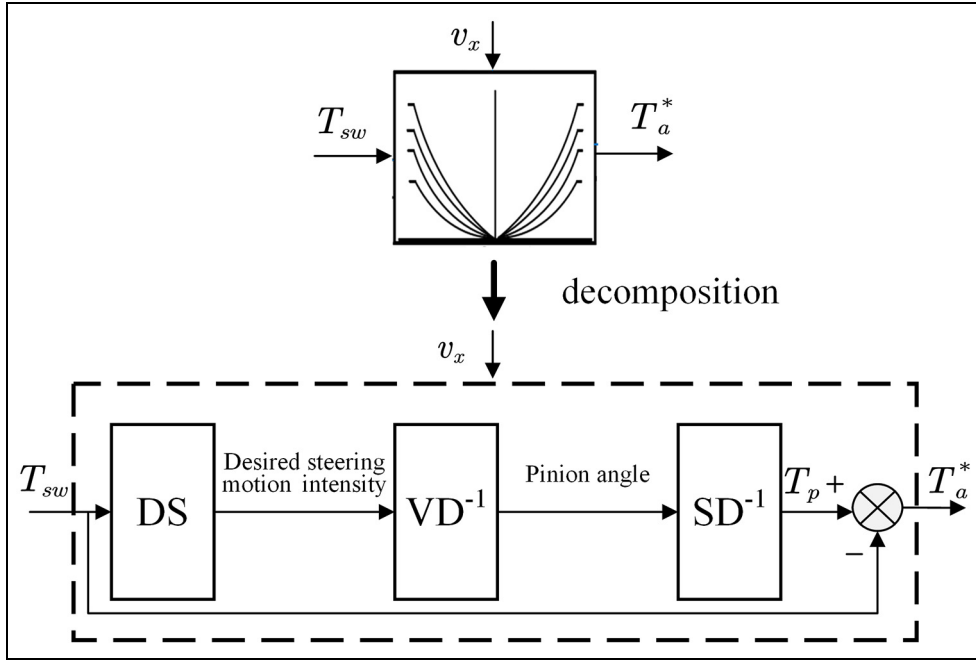


Figure 2. Diagram of the EPS assist characteristic curve decomposition and calculation method.

DS is short for driving style. VD^{-1} is short for the steady-state inverse characteristics of chassis dynamics. SD^{-1} is short for the steady-state inverse characteristics of steering system dynamics.

- (1) When the vehicle's longitudinal speed is low, the vehicle's steering dynamic response is not apparent, and even a substantial steering wheel angle change will not cause a significant change in lateral acceleration. It is difficult for the driver to perceive small changes in lateral acceleration, but drivers can perceive changes in the steering wheel angle. Therefore, the position of the steering wheel represents the steering motion intensity of the vehicle and indicates the driver's steering intention. The corresponding relationship between the driver's steering input torque and the steering wheel angle represents the driving style of the vehicle.
- (2) When the vehicle's longitudinal speed is high, the vehicle's steering dynamic response is obvious, and a small steering wheel angle change will result in a sizeable lateral acceleration change. The driver can perceive the change of lateral acceleration, but the perception of steering wheel angle change is fuzzy. Therefore, the lateral acceleration represents the steering motion intensity of the vehicle and indicates the driver's steering intention. The corresponding relationship between the driver's steering input torque and the lateral acceleration represents the driving style of the vehicle.

In summary, when the longitudinal speed is low, the driving style can be expressed by the correspondence between the driver's steering input torque and the steering wheel angle. When the longitudinal vehicle speed is high, the driving style can be expressed by the

corresponding relationship between the driver's steering input torque and the lateral acceleration.

In this paper, a parameterized driving style model was proposed. The boundary speed v_b to distinguish high and low speed was selected to be 30km/h. The driving style model below the boundary speed is shown in equation (6), and the driving style model above the boundary speed is shown in equation (7).

$$\delta_{sw}^* = \begin{cases} \text{sign}(T_{sw}) \min\left(\frac{-1}{f_1} \ln\left(\frac{1}{d} \left(c - \frac{|T_{sw}|}{1 + a \times v_x^b}\right)\right)\right), \delta_{swmax} & \dots, |T_{sw}| > (1 + a \times v_x^b) \times (c - d) \\ 0 & , |T_{sw}| \leq (1 + a \times v_x^b) \times (c - d) \end{cases} \quad (6)$$

$$a_y^* = \begin{cases} \text{sign}(T_{sw}) \min\left(\frac{-1}{f_2} \ln\left(\frac{1}{d} \left(c - \frac{|T_{sw}|}{1 + a \times v_x^b}\right)\right)\right), a_{ymax} & \dots, |T_{sw}| > (1 + a \times v_x^b) \times (c - d) \\ 0 & , |T_{sw}| \leq (1 + a \times v_x^b) \times (c - d) \end{cases} \quad (7)$$

where δ_{sw}^* and a_y^* are the driver's desired steering wheel angle and lateral acceleration, respectively, a and b are the speed factors, which indicate the effect of speed on the driver's desired steering motion intensity, c is the parking factor, indicating the driver's desired maximum torque when turning in parking maneuver, d is the dead zone factor, which together with c determines the size of the unassisted zone in parking maneuver, f_1 and f_2 are torque perception factors, indicating the sensitivity of δ_{sw}^* and a_y^* to steering wheel torque, δ_{swmax} and a_{ymax} are the maximum achievable steering wheel angle and lateral acceleration, respectively.

Table 1. The ideal steering wheel effort characteristics when $v_x = 5 \text{ km/h N} \cdot \text{m}$.

δ_{sw}	$\pi/3$	$2\pi/3$	π	2π
T_{sw}	1.66	1.77	1.79	1.80

Table 2. The ideal steering wheel effort characteristics when $v_x = 40 \text{ km/h N} \cdot \text{m}$.

a_y	0.25	0.35	0.45	0.5
T_{sw}	2.25	2.35	2.43	2.53

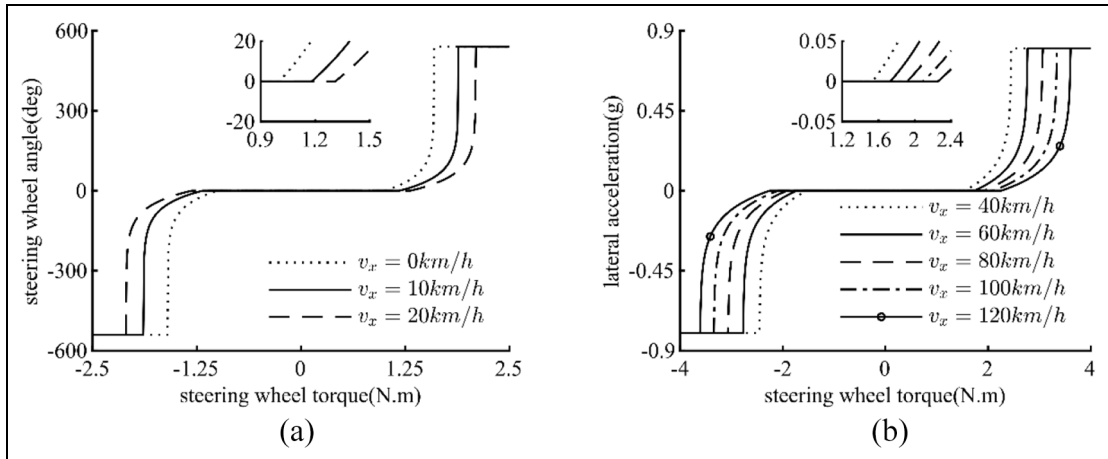
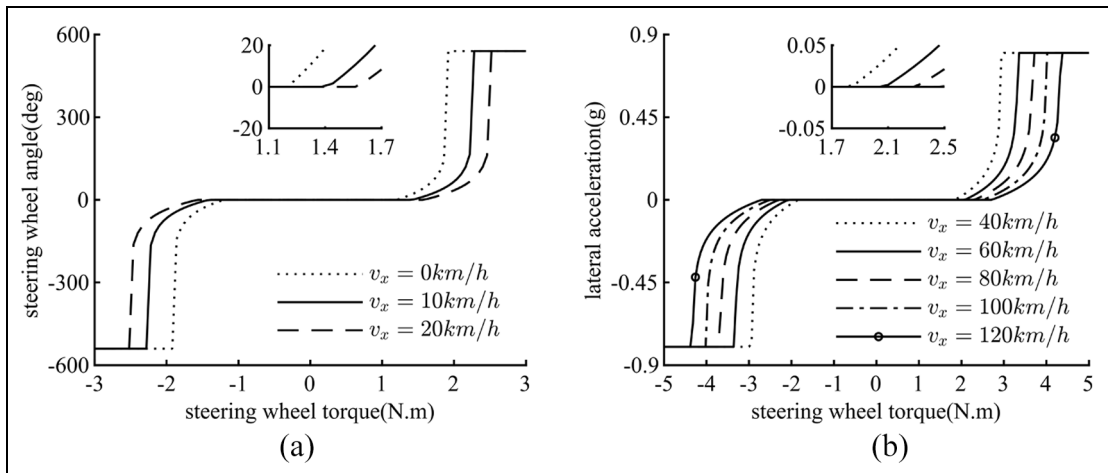
However, at present, there is no consensus on the ideal driving style. The desired driving style of consumers in different regions, genders, and ages are not the same. This paper referred to the research results of the ideal steering wheel effort characteristics of drivers in southwest China,¹⁷ as shown in Tables 1 and 2, and designed the desired driving style for drivers in southwest China.

The parameters of the driving style model were identified according to the data shown in Tables 1 and 2, and the final comfortable driving style was shown in Figure 3. The torque data shown in Tables 1 and 2 were enlarged by 1.2 times, and they were regarded as ideal sporty steering wheel effort characteristics, and the final sporty driving style was shown in Figure 4.

Calibration of steady-state inverse characteristics of chassis dynamics

The steady-state inverse characteristic of chassis dynamics is the corresponding relationship between the steering motion intensity and the pinion angle, which is the inherent characteristic of the vehicle. With the change of longitudinal speed, the steering motion intensity reflecting the driver's steering intention changes, and the steady-state inverse characteristic of chassis dynamics also changes accordingly, as follows:

- (1) When the longitudinal speed is less than the boundary speed, the steering motion intensity

**Figure 3.** Comfortable driving style: (a) $v_x < v_b$ and (b) $v_x > v_b$.**Figure 4.** Sporty driving style: (a) $v_x < v_b$ and (b) $v_x > v_b$.

reflecting the driver's steering intention is the steering wheel angle, and the steady-state inverse characteristic of chassis dynamics is the corresponding relationship between the steering wheel angle and the pinion angle. At steady state, the two satisfy the relationship shown in equation (4).

- (2) When the longitudinal speed is higher than the boundary speed, the steering motion intensity reflecting the driver's steering intention is the lateral acceleration, and the steady-state inverse characteristic of chassis dynamics is the corresponding relationship between the lateral acceleration and the pinion angle. Considering that the equivalent lateral stiffness of the front and rear axle of a vehicle is unsuitable to be measured and the vehicle has strong nonlinear characteristics, equation (5) is difficult to accurately express the corresponding relation between the lateral acceleration and the pinion angle. Therefore, this paper measures the relation through simulation based on the ASCL vehicle quality dynamics model of Jilin University.

Experiment method

- (1) Test conditions. The setting of longitudinal speed: Six test longitudinal speeds are selected, which are 30, 40, 60, 80, 100, and 120 km/h; The setting of lateral acceleration: The lateral acceleration is between -0.6 and 0.6 g, with 0.1 g as the test interval.
- (2) Test procedure. Select a set of test longitudinal speed and test lateral acceleration. First, the driver controls the car to accelerate to the target speed, then slowly turns the steering wheel until the lateral acceleration reaches the target value. Record the lateral acceleration, steering wheel torque, and steering wheel angle at steady state. Repeat the above process until all set condition tests are completed.

According to the steering wheel torque and steering wheel angle at steady state, the pinion angle at steady state can be calculated from equation (4). Thus the original data of the pinion angle relative to the lateral acceleration is obtained, as shown in Figure 5. In order to fill the missing pinion angle corresponding to the sizeable lateral acceleration, the following formula is used to fit the original data.

$$\delta_p = (a \times v_x^b + c) \times a_y \quad (8)$$

where a , b , and c are fitting related parameters, indicating the influence of velocity on the lateral acceleration gain.

After the above fitting treatment, the complete steady-state inverse characteristics of the chassis dynamics above the boundary speed were obtained, as shown in Figure 5.

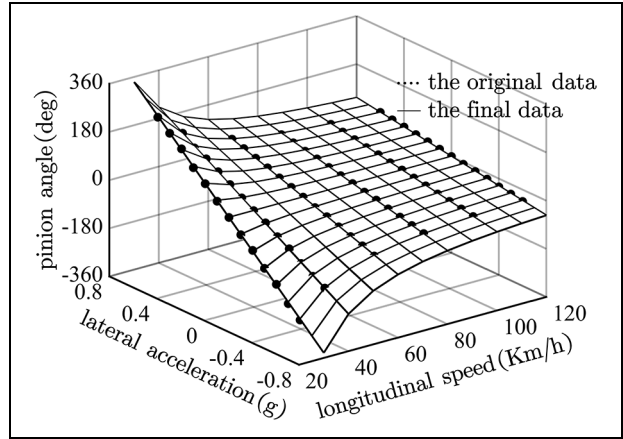


Figure 5. The steady-state inverse characteristics of chassis dynamics when $v_x > v_b$.

Calibration of steady-state inverse characteristics of steering system dynamics

The steady-state inverse dynamics of steering system dynamics is the relationship between the pinion angle and the equivalent steering resistance torque to the pinion when the driver turns outwards, which is the inherent characteristic of the vehicle. In fact, the steering resistance torque has visible nonlinear characteristics, which is produced by the lateral, longitudinal, vertical forces acting on the tire and the internal friction forces of the steering system. The equation (3) only considers the part caused by lateral force, so it is difficult to accurately describe the change of steering resistance torque during the driver's external steering. Therefore, this paper measures the steady-state inverse characteristics of steering system dynamics through simulation based on the ASCL vehicle quality dynamics model of Jilin University.

Experiment method

- (1) Test conditions. The setting of longitudinal speed: Eight test longitudinal speeds are selected, which are 0, 20, 40, 60, 80, 100, and 120 km/h; The setting of steering wheel angle: the steering wheel angle is between -520° and 520° , when the speed is less than the boundary speed, the test interval is 40° , when the speed is greater than or equal to the boundary speed and less than 80 km/h, the test interval is 20° , and when the speed is greater than or equal to 80 km/h, the test interval is 10° .
- (2) Test procedure. Close the assist system and select a set of test longitudinal speed and steering wheel angle. First, the driver controls the car to accelerate to the target speed, then slowly turns the steering wheel to the target angle or lateral acceleration higher than 0.65 g. If the steering wheel is turned to the target position, record the lateral acceleration, steering wheel torque and steering wheel angle at steady state. Repeat the above process until all set condition tests are completed.

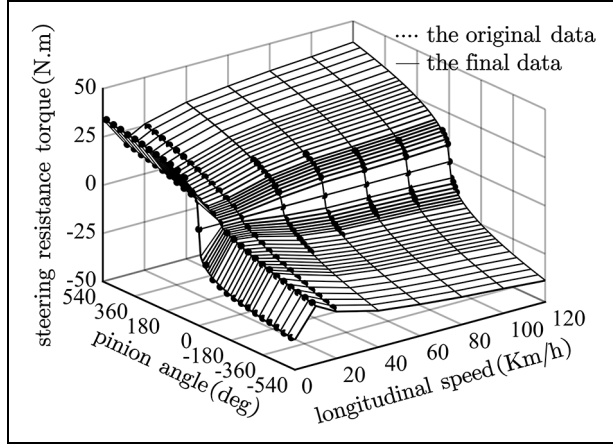


Figure 6. The steady-state inverse characteristics of the steering system.

When the assist system is closed, the assist torque is zero. From equation (1), it can be seen that the steering wheel torque and the steering resistance torque are equal at steady state. Thus the original data of the steering resistance torque relative to the pinion angle is obtained, as shown in Figure 6. In order to fill the missing steering resistance torque corresponding to the large pinion angle, the following formula is used to fit the original data at each speed.

$$T_p = (a \times |\delta_p|^b) \times \text{sign}(\delta_p) \quad (9)$$

where a and b are fitting related parameters, which have different values at different speeds.

After the above fitting treatment, the complete steady-state inverse characteristics of the steering system are obtained, as shown in Figure 6.

Calculation of EPS assist characteristic curve

The EPS assist characteristic curve can be calculated directly based on the designed driving style, the calibrated steady-state inverse characteristics of chassis dynamics and steering system dynamics.

The calculation method of the EPS assist characteristic curve is shown in Figure 2. First, according to the steering wheel torque and longitudinal speed signals, the driving style module is checked to determine the driver's desired steering motion intensity, when the longitudinal speed is less than the boundary speed, it is the steering wheel Angle; otherwise, it is the lateral acceleration. Secondly, the corresponding pinion angle is determined according to the steady-state inverse characteristic of chassis dynamics, and then according to the pinion angle and longitudinal speed, the steady-state inverse characteristics of steering system dynamics is checked to determine the steering resistance torque. Finally, the target assist torque can be obtained by subtracting the smaller of the steering wheel torque and the ideal steering wheel torque corresponding to the

maximum steering motion intensity from the steering resistance torque, as shown in the following formula:

$$T_a^* = \begin{cases} T_p - \text{sign}(T_{sw}) \times \min(|T_{sw}|, \dots \\ (1 + av_x^b)(c - de^{-f_2\delta_{sw\max}})) & , v_x < v_b \\ T_p - \text{sign}(T_{sw}) \times \min(|T_{sw}|, \dots \\ (1 + av_x^b)(c - de^{-f_2a_{y\max}})) & , v_x \geq v_b \end{cases} \quad (10)$$

Besides, the target assist torque is corrected according to equation (11) to ensure that the target assist torque and steering wheel torque are in the same direction.

$$T_a^* = \begin{cases} 0, T_{sw} \times T_a^* < 0 \\ T_a^*, T_{sw} \times T_a^* \geq 0 \end{cases} \quad (11)$$

Finally, the calculated comfortable and sporty EPS assist characteristic curve are shown in Figure 7.

Driving simulator virtual field test verification

To verify the effectiveness of the decomposition and calculation method of the EPS assist characteristic curve proposed in Section 3 to control steering feel, the driving simulator was taken as the test object as shown in Figure 8, and the calculated EPS assist characteristic curve was used for static steering efforts test, steering portability test and weave test.

Static steering efforts test

To verify the effectiveness of steering feel control during parking, the static steering efforts test was carried out. During the test, the longitudinal speed was zero, and the steering wheel was controlled by the steering robot to change slowly within the maximum angle range.

The relation curve between steering wheel angle and steering wheel torque of the static steering efforts test is shown in Figure 9. It can be seen that the steering wheel torque is less than 2 N.m in the full steering wheel angle range, and the relationship curve between steering wheel angle and steering wheel torque when the driver turns outside is consistent with the designed driving style whether the curve is a comfortable style or a sporty style. The results show that the EPS system adopting the assist characteristic curve calculated can significantly reduce the driver's pivot steering effort, improve steering portability, and effectively control the pivot steering feel when the driver turns outside, and realize the designed driving style.

Steering portability test

To verify the effectiveness of steering feel control at low speed, the steering portability test was carried out. For the test procedure, refer to the section on steering portability in the Chinese controllability and stability test procedure for automobiles.²⁰ During the test, the driver

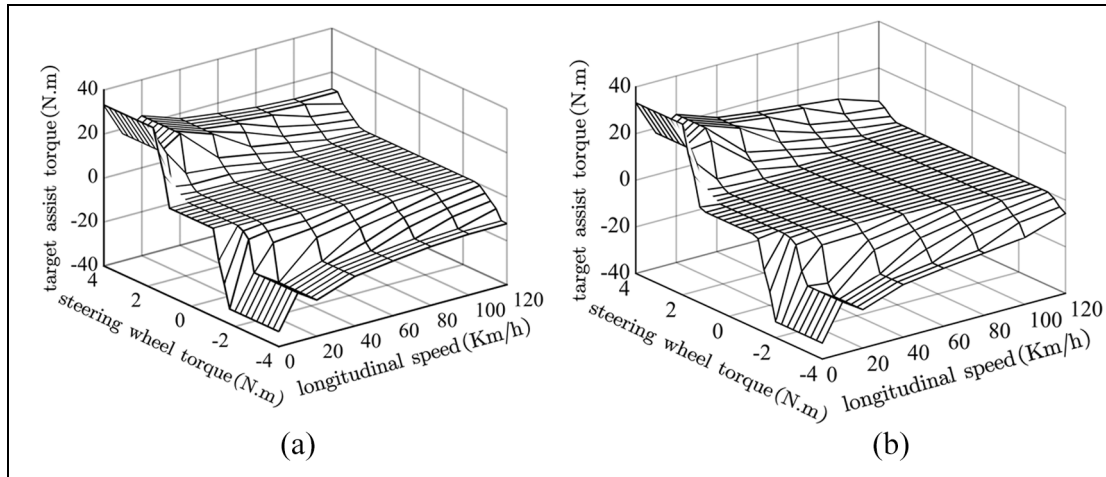


Figure 7. The EPS assist characteristic curve obtained by calculation: (a) comfortable style, and (b) sporty style.



Figure 8. ASCL driving simulator.

controlled the driving simulator and drove along the track of the double twist line on the field at a longitudinal speed of 10 ± 2 km/h.

The relation curve between steering wheel angle and steering wheel torque of the steering portability test is shown in Figure 10. It can be seen that the steering wheel torque in the steering process is about $2 \text{ N} \cdot \text{m}$, and the relationship curve between steering wheel angle and steering wheel torque when the driver turns outside is consistent with the designed driving style whether the curve is a comfortable style or a sporty style. The results show that the EPS system adopting assist characteristic curve calculated can ensure the steering portability at low speed, and effectively control the low-speed steering feel when the driver turns outside, and realize the designed driving style.

Weave test

To verify the effectiveness of steering feel control at high speed, the weave test was carried out. For the test procedure, refer to the section on the quantification of on-center handling in the Chinese controllability and

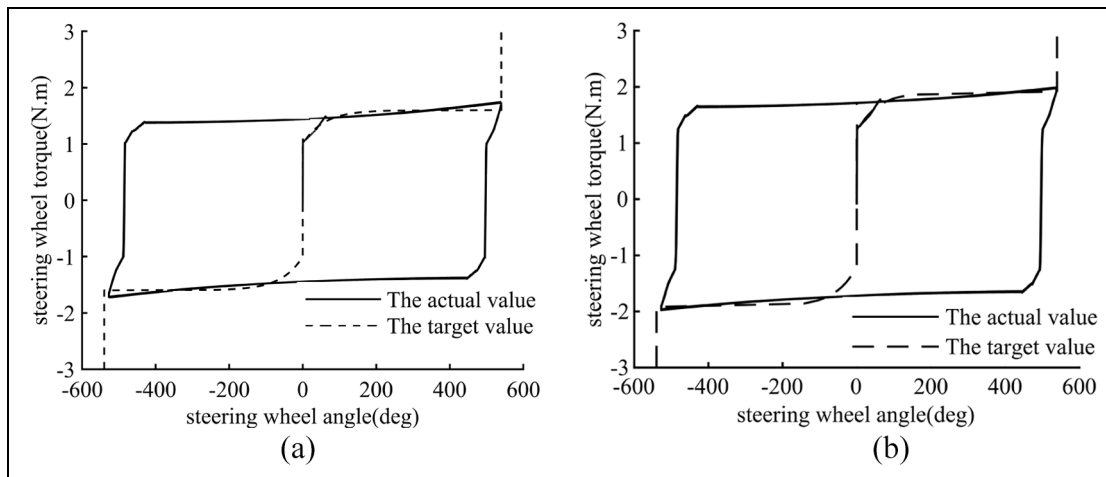


Figure 9. Relationship between steering wheel angle and steering wheel torque in the static steering efforts test: (a) comfortable style, and (b) sporty style.

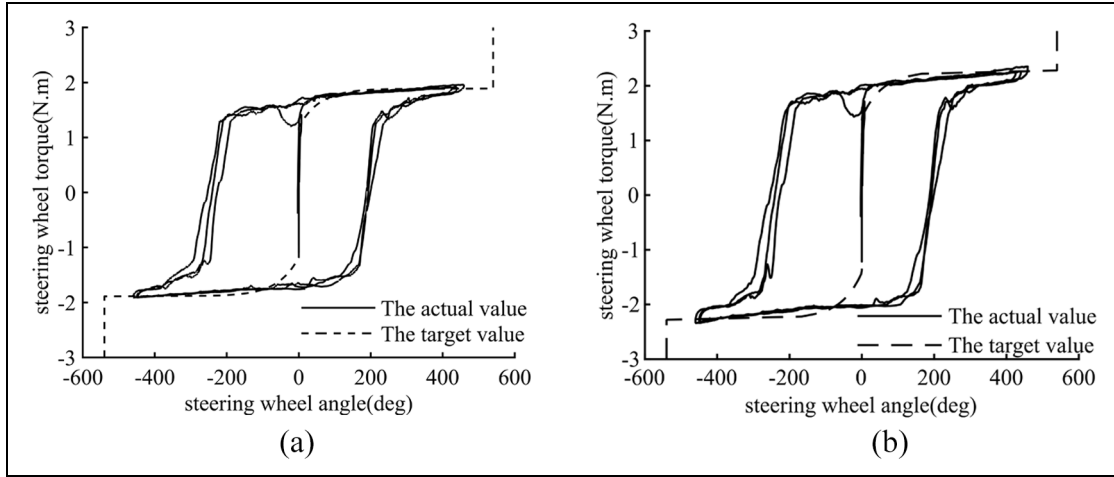


Figure 10. Relationship curve of steering wheel angle and torque in the steering portability test: (a) comfortable style and (b) sporty style.

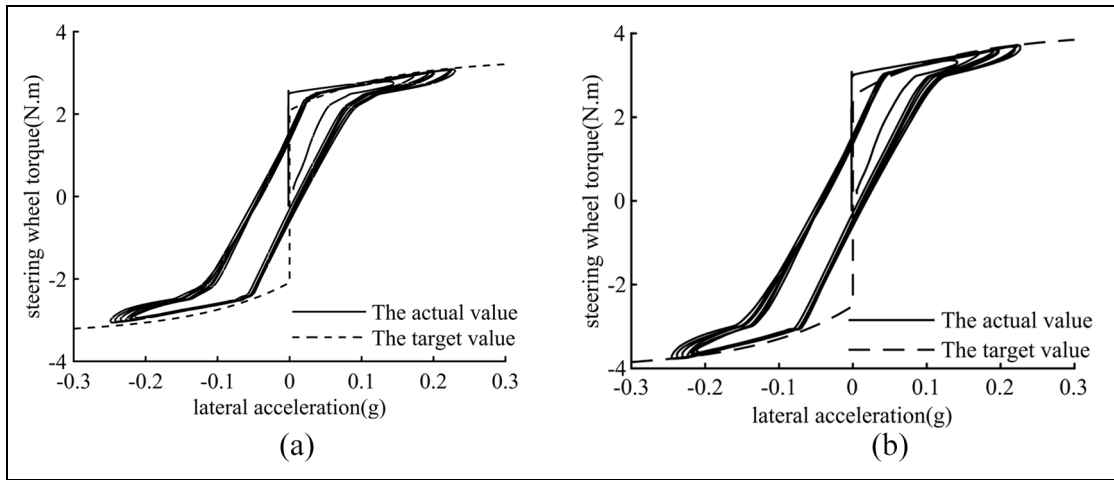


Figure 11. Relationship curve of lateral acceleration and steering wheel torque in the weave test: (a) comfortable style and (b) sporty style.

stability test procedure for automobiles.²⁰ During the test, the driver controlled the driving simulator and drove at a longitudinal speed of 100 ± 2 km/h, and controlled the steering wheel continuous sinusoidal steering at a frequency of 0.2 Hz, so that the lateral acceleration of the vehicle reached 2 m/s^2 .

The relation curve between lateral acceleration and steering wheel torque of the weave test is shown in Figure 11. It can be seen that the steering wheel torque is between 2 and $3 \text{ N} \cdot \text{m}$, and the relationship curve between lateral acceleration and steering wheel torque when the driver turns outside is consistent with the designed driving style whether the curve is a comfortable style or a sporty style. The results show that the EPS system adopting assist characteristic curve calculated can improve the stability of high-speed steering, ensure clear feedback of the road feel at high speed, and effectively control the high-speed steering feel when the driver turns outside, and realizes the designed driving style.

Conclusion

- (1) The mechanism of the EPS assist characteristic curve is clarified. It is found that the process of designing and adjusting the EPS assist characteristic curve is a process of adjusting the corresponding relationship between the steering wheel torque and the steering motion intensity based on considering vehicle dynamic characteristics, which provides a theoretical basis for its design and adjustment.
- (2) A decomposition and calculation method of the EPS assist characteristic curve is proposed, which takes the driver's desired steering motion intensity and the pinion angle position as intermediate variables. The EPS assist characteristic curve is decomposed into three parts: driving style, steady-state inverse characteristics of chassis dynamics, and steady-state inverse characteristics of steering system dynamics. The EPS assist characteristic

curve can be calculated directly based on the designed driving style, the calibrated steady-state inverse characteristics of chassis dynamics and steering system dynamics. The decomposition and calculation method proposed in this paper decouples the driver's steering intention and vehicle dynamics characteristics coupled in the EPS assist characteristic curve and eliminates the influence of chassis dynamics and steering system dynamics on steering feel. The steering feel can be directly controlled by designing the driving style, which lays the foundation for satisfying the driver's personalized steering needs.

- (3) The experimental tests are conducted driven by the driving simulator. The test results show that the EPS adopting assist characteristic curve calculated in this paper can realize the designed driving style and provide consistent and controllable steering feel on the premise of ensuring steering lightness at low speed and steering stability at high speed.
- (4) In the future, the EPS double closed-loop control strategy based on the decomposition method of the EPS assist characteristic curve proposed in this paper will be studied. Through the closed-loop control of steering motion intensity and pinion angle, the requirements for the calibration accuracy of chassis dynamics and steering system dynamics will be reduced, the influence of dynamic characteristics like empty trip and damping will be eliminated, the designed driving style will be realized throughout the steering process regardless of acceleration or deceleration.

Declaration of conflicting interests

The author(s) declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Funding

The author(s) disclosed receipt of the following financial support for the research, authorship, and/or publication of this article: This work was supported by the National Key R&D Program of China (No. 2018YFB1502702).

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